

1. Simple Chat Application

Project Title

Simple Chat Application

Description

A Simple Chat Application is a communication system that allows users to send and receive messages in real time. The application will provide basic user registration, login, one-to-one chatting, and message management features.

Objectives

1. To provide a simple platform for communication between users.
2. To implement user registration and authentication.
3. To allow users to send and receive text messages.
4. To store and retrieve chat messages using file handling.
5. To understand basic client-server/API-based communication.

Tools and Technology

- C++ / Web Technologies
- HTML, CSS, JavaScript, if web-based
- File handling for storing users and messages
- Existing APIs for communication
- IDE such as Visual Studio Code / Code::Blocks

Manifestation / Functionalities

The application should include:

User Module

- User registration
- User login/logout
- User profile
- Change password

Chat Module

- View available users
- Select a user for chatting
- Send text messages
- Receive messages
- Display messages with sender and time
- View previous conversations
- Clear/delete conversation

File Storage

- Store user information in files
- Store chat messages in files
- Read previous messages when a conversation is opened

Additional Features

- Search users
- Online/offline status
- Basic input validation
- Logout functionality

Expected Outcomes

Students will be able to:

- Build a basic communication application.
 - Understand client-server/API communication.
 - Implement file-based data storage.
 - Develop user authentication and messaging functionality.
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2. Online Voting System

Project Title

Online Voting System

Description

An Online Voting System is an application that allows registered users to cast their votes electronically. The system will manage voters, candidates, voting activities, and election results while ensuring that each voter can vote only once.

Objectives

1. To provide a simple electronic voting system.
2. To manage voter and candidate information.
3. To prevent duplicate voting.
4. To automatically calculate election results.
5. To maintain voting records using file handling.

Tools and Technology

- C++ / Web Technologies
- HTML, CSS, JavaScript

- File handling for data storage
- Existing APIs where required

Manifestation / Functionalities

Admin Module

- Admin login
- Add candidates
- Update candidate information
- Delete candidates
- View candidate list
- Create/manage election
- View voting results

Voter Module

- Voter registration
- Voter login
- Voter verification
- View candidates
- View candidate details
- Cast vote
- Prevent multiple voting
- View voting confirmation

Result Module

- Count votes automatically
- Display votes received by each candidate
- Display winner
- Generate election summary

File Storage

- Voter information
- Candidate information
- Voting status
- Vote records
- Election results

Expected Outcomes

Students will understand authentication, validation, file handling, vote counting, and basic election-management functionality.

3. Parking Management System

Project Title

Parking Management System

Description

A Parking Management System is designed to manage vehicles entering and leaving a parking area. It will maintain parking slots, vehicle details, entry/exit times, and parking charges.

Objectives

1. To manage parking spaces efficiently.
2. To record vehicle entry and exit.
3. To calculate parking charges.
4. To display available and occupied slots.
5. To maintain parking records using files.

Tools and Technology

- C++ / Web Technologies
- HTML, CSS, JavaScript
- File handling
- Existing APIs if required

Manifestation / Functionalities

Vehicle Management

- Vehicle entry
- Vehicle number registration
- Vehicle type selection
- Owner details
- Generate parking/token number

Parking Slot Management

- Display available slots
- Allocate parking slot
- Mark slot as occupied
- Release slot after vehicle exit
- Display occupied slots

Entry/Exit Management

- Record entry time
- Record exit time
- Calculate parking duration
- Calculate parking fee

- Generate parking receipt

Admin Features

- View all parked vehicles
- Search vehicle by number
- View available slots
- View daily parking records

File Storage

- Vehicle records
- Slot information
- Entry/exit records
- Payment records

Expected Outcomes

Students will learn record management, file handling, searching, time-based calculation, and basic resource management.

4. E-Commerce Platform

Project Title

E-Commerce Platform

Description

An E-Commerce Platform allows users to browse products, search products, add items to a cart, place orders, and view order details. An administrator can manage products and orders.

Objectives

1. To develop an online product purchasing platform.
2. To implement product and category management.
3. To provide shopping-cart functionality.
4. To manage customer orders.
5. To understand basic e-commerce workflow.

Tools and Technology

- C++ / Web Technologies
- HTML, CSS, JavaScript
- File handling
- Existing APIs for required services

Manifestation / Functionalities

Customer Module

- Registration/login
- Browse products
- Search products
- Filter products by category
- View product details
- Add product to cart
- Update quantity
- Remove product from cart
- View total price
- Place order
- View order history

Admin Module

- Admin login
- Add products
- Update products
- Delete products
- Manage categories
- View customers
- View orders
- Update order status

Cart Module

- Add items
- Remove items
- Change quantity
- Calculate subtotal
- Calculate total amount

File Storage

- User records
- Product records
- Cart information
- Order records
- Category information

Expected Outcomes

Students will understand the complete basic shopping workflow from product selection to order placement.

5. Library Management System with Fine Calculation

Project Title

Library Management System with Fine Calculation

Description

A Library Management System manages books, students/members, book issue and return operations, and calculates fines for late returns.

Objectives

1. To manage library books and members.
2. To automate book issue and return.
3. To calculate late-return fines.
4. To track available and issued books.
5. To maintain library records using files.

Tools and Technology

- C++ / Web Technologies
- HTML, CSS, JavaScript
- File handling
- Date/time functions

Manifestation / Functionalities

Book Management

- Add book
- Update book
- Delete book
- Search book
- View available books
- View issued books

Member Management

- Add member/student
- Update member
- Delete member
- Search member

Issue Module

- Issue book
- Record issue date
- Record expected return date
- Check book availability

Return Module

- Return book
- Record return date
- Calculate number of late days
- Calculate fine
- Store payment/fine status

Fine Calculation

- Define fine per day
- Calculate total late days
- Calculate total fine automatically
- Display fine on return

Expected Outcomes

Students will learn CRUD operations, file handling, date calculations, searching, and automated fine calculation.

6. Mini Compiler

Project Title

Mini Compiler

Description

A Mini Compiler is a simplified compiler project that demonstrates the basic stages of language processing. It can accept a small set of programming statements and perform lexical analysis, syntax checking, and basic error detection.

Objectives

1. To understand the basic working of a compiler.
2. To identify tokens from source code.
3. To perform basic syntax checking.
4. To detect common programming errors.
5. To demonstrate different compiler phases.

Tools and Technology

- C++
- File handling
- String processing
- Data structures

Manifestation / Functionalities

Input Module

- Accept source code
- Read source code from a file
- Display source code

Lexical Analysis

- Identify keywords
- Identify identifiers
- Identify operators
- Identify constants
- Identify delimiters
- Generate token list

Syntax Analysis

- Check basic syntax rules
- Check brackets
- Check statement termination
- Check basic expression structure

Error Detection

- Invalid identifier
- Unknown symbol
- Missing semicolon
- Unmatched brackets
- Invalid expression

Output

- Display tokens
- Display detected errors
- Display successful syntax-check message

Expected Outcomes

Students will gain an understanding of lexical analysis, syntax analysis, tokens, parsing concepts, and error detection.

7. Weather Forecasting System

Project Title

Weather Forecasting System

Description

A Weather Forecasting System provides weather information for a selected location. The system can obtain weather information through an available weather API and display temperature, humidity, wind, and weather conditions.

Objectives

1. To provide weather information for different locations.
2. To understand API-based data retrieval.
3. To display weather information in a user-friendly format.
4. To store recent searches using files.

Tools and Technology

- C++ / Web Technologies
- HTML, CSS, JavaScript
- Weather API
- File handling
- JSON/API data processing

Manifestation / Functionalities

Location Module

- Enter city/location
- Search location
- Display selected location

Weather Module

- Current temperature
- Weather condition
- Humidity
- Wind speed
- Minimum/maximum temperature where API supports it
- Weather icon where available

Forecast Module

- Display upcoming forecast
- Display date-wise weather information

- Display temperature and conditions

History

- Store recently searched locations
- View previous searches
- Clear search history

Expected Outcomes

Students will learn API integration, JSON/data processing, file handling, and dynamic display of external data.

8. Health and Fitness Tracker

Project Title

Health and Fitness Tracker

Description

A Health and Fitness Tracker helps users record daily physical activities such as steps, exercise, water intake, calories, and weight. The application provides summaries to help users monitor their fitness activities.

Objectives

1. To record daily fitness activities.
2. To calculate basic fitness statistics.
3. To provide daily and weekly summaries.
4. To maintain personal activity records using files.

Tools and Technology

- C++ / Web Technologies
- HTML, CSS, JavaScript
- File handling
- Basic mathematical calculations

Manifestation / Functionalities

User Profile

- Name
- Age
- Height

- Weight
- Fitness goal

Activity Tracking

- Steps
- Exercise type
- Exercise duration
- Water intake
- Calories consumed/burned
- Daily weight

Calculation

- BMI calculation
- Daily activity summary
- Weekly activity summary
- Exercise duration calculation

Progress

- View daily records
- View weekly records
- Compare progress
- Display achievement/goal status

File Storage

- User profile
- Daily activity
- Exercise records
- Weight records

Expected Outcomes

Students will learn file handling, calculations, record management, and basic data presentation.

9. Bus Reservation System

Project Title

Bus Reservation System

Description

A Bus Reservation System allows users to search available buses, select seats, make reservations, and view or cancel their bookings.

Objectives

1. To provide an online bus reservation facility.
2. To manage buses and seats.
3. To automate seat booking.
4. To prevent duplicate seat booking.
5. To maintain reservation records using files.

Tools and Technology

- C++ / Web Technologies
- HTML, CSS, JavaScript
- File handling

Manifestation / Functionalities

Search Module

- Enter source
- Enter destination
- Select journey date
- Display available buses

Bus Details

- Bus number
- Bus name
- Source
- Destination
- Departure time
- Arrival time
- Fare
- Available seats

Seat Module

- Display seat layout
- Show available seats
- Show booked seats
- Select seat
- Prevent duplicate booking

Reservation

- Passenger details
- Seat selection
- Fare calculation

- Confirm booking
- Generate ticket/booking ID

Cancellation

- Search booking
- Cancel ticket
- Update seat availability

Expected Outcomes

Students will learn booking workflows, seat management, file handling, searching, and transaction-style processing.

10. Movie Ticket Booking System

Project Title

Movie Ticket Booking System

Description

A Movie Ticket Booking System allows users to view movies, theatres, show timings, available seats, and book tickets for selected shows.

Objectives

1. To provide an online movie-ticket booking facility.
2. To manage movies, theatres, shows, and seats.
3. To implement seat selection.
4. To calculate ticket costs.
5. To maintain booking records.

Tools and Technology

- C++ / Web Technologies
- HTML, CSS, JavaScript
- File handling

Manifestation / Functionalities

Movie Module

- Display movies
- Movie details
- Search movies

- Movie category/language

Show Module

- Theatre selection
- Show date
- Show timing
- Ticket price

Seat Module

- Display seat layout
- Available seats
- Booked seats
- Select multiple seats

Booking

- Customer details
- Seat selection
- Calculate total price
- Generate booking ID
- Generate ticket

Cancellation

- Search booking
- Cancel ticket
- Update seat availability

Admin

- Add/update/delete movies
- Manage theatres
- Manage show timings
- View bookings

Expected Outcomes

Students will understand reservation systems, seat management, file handling, and ticket generation.

11. Flight Reservation System

Project Title

Flight Reservation System

Description

A Flight Reservation System allows users to search flights, view flight details, select seats, reserve tickets, and manage their bookings.

Objectives

1. To provide a basic flight reservation facility.
2. To manage flight schedules.
3. To implement passenger and seat management.
4. To calculate ticket fares.
5. To maintain booking information using files.

Tools and Technology

- C++ / Web Technologies
- HTML, CSS, JavaScript
- File handling

Manifestation / Functionalities

Flight Search

- Source
- Destination
- Travel date
- Display available flights

Flight Details

- Flight number
- Airline name
- Departure time
- Arrival time
- Duration
- Fare
- Available seats

Passenger Module

- Passenger name
- Age
- Gender
- Contact details
- ID information

Seat Reservation

- Display seats

- Select seat
- Check seat availability
- Prevent duplicate seat booking

Booking

- Calculate fare
- Confirm reservation
- Generate booking/PNR number
- Display ticket

Cancellation

- Search booking
- Cancel reservation
- Update seat availability

Expected Outcomes

Students will learn search, reservation, seat management, file handling, and ticket-generation concepts.

12. Currency Converter

Project Title

Currency Converter

Description

A Currency Converter allows users to convert an amount from one currency to another using exchange-rate information. The system can use an available currency API or predefined exchange rates.

Objectives

1. To provide quick currency conversion.
2. To understand API-based exchange-rate retrieval.
3. To perform accurate conversion calculations.
4. To maintain recent conversion history.

Tools and Technology

- C++ / Web Technologies
- HTML, CSS, JavaScript
- Currency API

- File handling

Manifestation / Functionalities

Currency Selection

- Select source currency
- Select target currency
- Enter amount

Conversion

- Retrieve exchange rate
- Convert amount
- Display exchange rate
- Display converted amount

Additional Features

- Swap currencies
- Support multiple currencies
- Display last updated rate when API provides it
- Conversion history
- Clear history

Expected Outcomes

Students will learn API integration, mathematical calculations, JSON/data processing, and file handling.

13. Online Shopping Cart System

Project Title

Online Shopping Cart System

Description

An Online Shopping Cart System allows customers to browse products and manage selected products in a shopping cart. The system focuses mainly on product selection, cart management, price calculation, and order placement.

Objectives

1. To implement shopping-cart functionality.
2. To manage products and prices.

3. To calculate order totals.
4. To provide basic order placement.
5. To store cart and order information using files.

Tools and Technology

- C++ / Web Technologies
- HTML, CSS, JavaScript
- File handling

Manifestation / Functionalities

Product Module

- Display products
- Product details
- Search products
- Filter by category
- Display price and stock

Shopping Cart

- Add product
- Remove product
- Increase/decrease quantity
- View cart
- Calculate subtotal
- Calculate total amount

Checkout

- Customer details
- Delivery address
- Order summary
- Confirm order
- Generate order ID

Order Management

- View order history
- View order details
- Cancel order

Admin

- Add product
- Update product
- Delete product
- Update stock
- View orders

Expected Outcomes

Students will understand shopping-cart logic, CRUD operations, calculations, file handling, and order processing.

14. Student Result Management System

Project Title

Student Result Management System

Description

A Student Result Management System manages student academic information, subject marks, total marks, percentage, grade, and pass/fail status. It provides an organized method for entering and viewing examination results.

Objectives

1. To manage student academic records.
2. To calculate total marks and percentage.
3. To automatically assign grades.
4. To generate student result reports.
5. To store student information using files.

Tools and Technology

- C++ / Web Technologies
- HTML, CSS, JavaScript
- File handling
- Basic mathematical calculations

Manifestation / Functionalities

Student Module

- Add student
- Student ID/roll number
- Name
- Course/class
- Semester
- Contact information

Subject/Marks Module

- Add subjects

- Enter marks
- Update marks
- View subject-wise marks
- Validate marks

Result Calculation

- Calculate total marks
- Calculate percentage
- Calculate grade
- Determine pass/fail
- Calculate GPA/CGPA if required by the project

Result Display

- Search student by roll number
- Display complete result
- Generate result summary
- Print/download result where applicable

Admin/Teacher

- Add/update student
- Enter marks
- Modify marks
- View all results
- Search results

Expected Outcomes

Students will learn file-based record management, calculations, searching, validation, and report generation.

15. To-Do List Application

Project Title

To-Do List Application

Description

A To-Do List Application helps users create and manage daily tasks. Users can add tasks, assign priorities or deadlines, mark tasks as completed, and remove completed or unwanted tasks.

Objectives

1. To provide a simple task-management application.
2. To allow users to create and manage tasks.
3. To track completed and pending tasks.
4. To store tasks using file handling.
5. To provide a simple and user-friendly interface.

Tools and Technology

- C++ / Web Technologies
- HTML, CSS, JavaScript
- File handling
- Date/time functions

Manifestation / Functionalities

Task Management

- Add task
- Edit task
- Delete task
- Mark task as completed
- Mark task as pending
- View task details

Task Information

- Task title
- Description
- Priority
- Due date
- Status

Task Display

- View all tasks
- View completed tasks
- View pending tasks
- View overdue tasks
- Search tasks
- Filter by priority/status

File Storage

- Save tasks
- Load tasks when application starts
- Update task records
- Delete task records

Expected Outcomes

Students will learn file handling, CRUD operations, searching, filtering, date handling, and basic application design.

16. Chatbot

Project Title

Chatbot

Description

A Chatbot is an application that interacts with users through text-based conversations. The chatbot will receive user questions, identify predefined keywords or patterns, and provide appropriate responses.

Objectives

1. To develop a basic conversational application.
2. To understand input processing and pattern matching.
3. To provide automated responses to common questions.
4. To maintain basic conversation history.
5. To understand the fundamentals of chatbot development.

Tools and Technology

- C++ / Web Technologies
- HTML, CSS, JavaScript
- File handling
- String processing
- Existing APIs, if an API-based chatbot is selected

Manifestation / Functionalities

Conversation Module

- Display chat interface
- Accept user message
- Process user input
- Generate response
- Display chatbot response

Basic Intelligence

- Keyword matching
- Predefined questions and answers
- Greeting detection

- Goodbye detection
- Frequently asked questions
- Unknown-question response

Knowledge/File Module

- Store questions and answers in a file
- Read responses from the file
- Add new questions/responses
- Update existing responses

Conversation History

- Store previous messages
- Display current conversation
- Clear conversation

Optional API Feature

- Connect to an existing chatbot/AI API
- Send user input to API
- Receive response
- Display API response

Expected Outcomes

Students will learn string processing, pattern matching, file handling, API integration, and basic conversational-system development.